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Road Detection Network Based on Anti-Disturbance and Variable-Scale Spatial Context Detector

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ABSTRACT Road detection plays a critical role in the application of smart transportation. The performance of the mainstream methods such as PSPNet (Pyramid Scene Parsing Network), DeepLab V3, FCRN (Fully Convolutional Residual Networks) still suffers from uncertain disturbances of surface abrasion buildings, pedestrians, and variation of illumination like tree-shadow. The extracted features are vulnerable to extra-disturbance, and non-local spatial-context information has not been fully utilized. In this paper, a detector based on anti-disturbance and variable-scale spatial context features (AVD) is proposed: the training of the multi-layer features of the detector is always taken under the imposing of fake-feature-disturbance from an independent generator, which is trained to exacerbate the detector errors and the mistakes of feature discriminator. The detector is prepared to be immune from the fake-feature-disturbance, and the discriminator is trained to distinguish the differences between the non-interference features and disturbing features. We also designed a novel variable-scale spatial context module to enhance the richness performance of the extracted features. And a soft connection link is bridged between the low and high feature layers. The detection experiments on the Munich road dataset and urban road dataset show that AVD is better than all the mainstream above methods. Our method increases the accuracy by 3% on the Munich remote sensing dataset and 0.4% on the urban road dataset. Our code and datasets are available at https://github.com/Ding-Q/AVD for download.

INDEX TERMS Object detection, convolutional neural network, anti-disturbance, generative adversarial network, road detection, remote sensing road detection.

I. INTRODUCTION

Road detection is an essential task in the object detection of intelligent driving. By detecting lanes and areas on the roads, vehicles can achieve smart assisted driving, such as lane-keeping and lane departure assist systems. Traditional road detection [1], [2] mainly relied on the combination of setting ROI (Region of Interest) area and edge detection. These methods suffered from low accuracy and instability, and semantic segmentation methods on road detection based on deep learning FCN [3], DeepLab [6], PSPNet [7] have substituted these methods in recent years. Compared with the traditional method, the semantic segmentation method based on deep learning can obtain more advanced semantic information to express the scene in the image. However, the detection accuracy is still affected by some complex environmental disturbances such as surface abrasion caused by heavy vehicles, occlusion by water, garbage, etc., and influenced by the scarcity of the research works on utilizing global spatial semantic information. Therefore, it is crucial to improve the detection network feature performances on the anti-disturbance and extraction of global spatial context.

The mainstream semantic segmentation methods include: FCN proposed by Long [3], which replaces the full connection part of VGGNet [4] with transposed convolution, establishes a fully convolutional neural network, and completes the detection of pixels in the dense image classification, achieves the end-to-end learning. The FCRN (proposed by Laina *et al.* [5]) introduced a novel residual structure that used small-scale convolution kernels instead of large ones

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in up-sampling. DeepLab model (by Liang-Chieh *et al.* [6]) combined the Atrous Convolution with multi-scale pooling layers to obtain higher detection accuracy. The PSPNet model (by Zhao *et al.* [7]) concatenated spatial pyramid pooling layers to achieve multi-scale semantic feature fusion. The features of local and global clues together make the final prediction more reliable.

All the above semantic segmentation methods do not have the function of anti-disturbance. The traditional methods of directly adding interference items are Gaussian blur or salt-and-pepper noise, apt to image distortion. To this end, this paper proposes to superimpose the fake features generated by the jammer upon the feature layer of the network. It trains the jammer to influence the accuracy of the detection network as much as possible while ensuring the mixed features can pass the authenticity discrimination. Meanwhile, the detection net is trained to overcome the feature disturbances. Therefore, the jammer, the detection network, and the feature discriminator form generative adversarial relationships.

The generative adversarial relationship between generator and discriminator was first revealed by Goodfellow [8] in Generative Adversarial Networks (GAN) in 2014 and used to improve the quality of the synthesized images by a generator from 1D noise vectors. DCGAN [9] utilized shared convolution kernels instead of fully connected connections. Pix2Pix [10] combined 2D label information with 2D noise sample to obtain conditional generating pictures. StyleGAN [11] used a style modulator to generate style signals and inject them into the middle layer of the generator to achieve style-transferred images.

Based on the above research works, a fake feature generator is designed, which takes images and 2D noise as inputs, outputs fake feature items. Then fake feature items are pumped into the intermediate feature layers of the encoder in the detector network.

The mainstream semantic segmentation methods have not fully explored the potential of spatial context information, which plays a vital role in road detection. One most important pattern of spatial context information is the self-attention [12] module, which extracts the spatial context features by building spatial relationships among all pixel pairs via matrix multiplications. It was first introduced into the image handling process by SAGAN [13] to improve the quality of the generated images. Fu *et al.* [14] introduced a dual attention mechanism network (DANet) to combine local features with global dependencies to obtain more accurate segmentation results. However, there are two problems with the above self-attention mechanisms: 1). The module has relatively few trainable parameters. 2). When handling large images, the pixel matrix occupies huge GPU memory, limiting its application opportunity.

The traditional skip connection between encoder and decoder has been first proposed by the U-Net [15] network, which bridges the low and high layers to enhance the training efficiency. It has been widely used in semantic segmentation methods [16], [17]. However, such rigid skip connection ignores the incompatibility between different layers, which affects the improvement of training accuracy.

Against the shortcomings of the above methods, in this paper, a detector based on anti-disturbance and variable-scale spatial context features (AVD) is proposed: in the encoder part, the generated disturbance terms from a pre-designed jammer are imposed upon the extracted features to enhance the anti-disturbance robustness of our detector. In the decoder part, a novel variable-scale spatial context module is introduced and replicated between the layers of the detector, which increases more trainable parameters and reduces the GPU memory by variable-scale matrices, between the encoder and decoder, a soft skip connection is designed to avoid the compatibility problem of traditional rigid skip connections and improve training efficiency.

Our AVD network combines multi-anti-disturbance, variable-scale spatial context, and soft skip connection and gets a positive experimental result on two road datasets. The detection accuracy of Munich dataset has been improved by 2%, 1.5%, 0.5% and 3% via adding multi-anti-disturbance method, variable-scale spatial context module, soft skip connection and them all. The accuracy of the urban road dataset has been increased by 0.22%, 0.18%, 0.1% and 0.4% via adding the multi-anti-disturbance method, variable-scale spatial context module, soft skip connection and them all.

The main contributions of this paper are as follows: (1) we propose an anti-disturbance detection network, enhance its anti-disturbance performance by imposing feature-disturbance into the encoder. (2) A novel variable-scale spatial context module is designed to increase trainable parameters and reduce the spatial scale for large images. (3) A soft skip connection is introduced to improve the compatibility among different feature layers.

II. RELATED WORKS

A. FCRN

Fully Convolutional Residual Networks referred to as FCRN. Its author designs a new residual block structure (Fast Up-Projection), using four kinds of small-scale convolution kernels instead of large ones in up-sampling. On the one hand, it can reduce the checkerboard effect, on the other hand it can preserve edge information as much as possible. The author uses reverse Huber (*berHu*) [18] as the loss function and obtains a better result. The formula is as shown in eq. (1):

$$\mathcal{B}(x) = \begin{cases} |x|, & |x| \le c \\ \dfrac{x^2 + c^2}{2c}, & |x| > c \end{cases} \tag{1}$$

When x is between $[-c, c]$, the loss function is the L1 norm. Otherwise it is the L2 norm.

B. GAN

The Generative Adversarial Network is abbreviated as GAN, which is an unsupervised deep learning model composed of two networks: G (generator) and D (discriminator). The process of GAN is: G takes random Gaussian noise z as

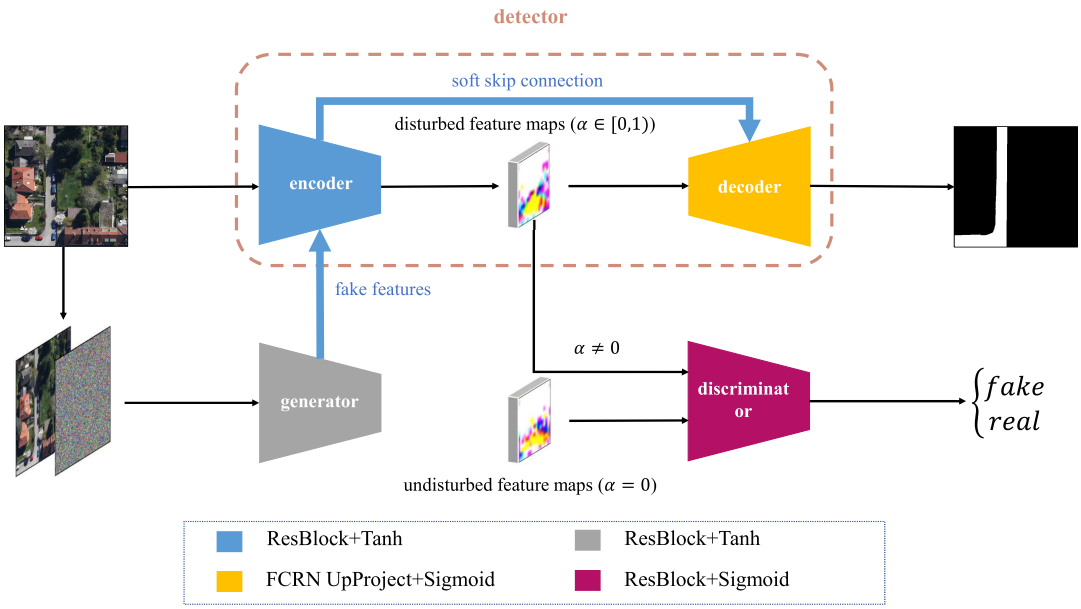

FIGURE 1. AVD network architecture. The encoder takes the road image as input and outputs the feature map of this image. The Gaussian noise and the image are fed simultaneously to the generator for producing fake features, which are proportionally imposed onto the middle layers of the encoder by $\alpha * \textit{fake features} + (1-\alpha) * \textit{real features}$, $0 \leq \alpha < 1$. We manually set the scale coefficient α to change the proportion of the fake feature in the middle layers of the encoder. The decoder accepts the disturbed features and real features then outputs the detected results. The discriminator distinguishes disturbed features from real features.

input, generates an image $G(z)$. And D takes the real image x and the generated image $G(z)$ as inputs. D does a two-class classification of x and $G(z)$ to detect the real and fake of the image. The output of D is 0 or 1, 0 represents a generated image, and 1 represents a real image. The objective function can be expressed as formula (2):

$$\min_G \max_D V(G,D) = E_{x\sim P_{\text{data}}} \log D(x) + E_{z\sim P_z} \log(1 - D(G(z))) \quad (2)$$

The goal of the discriminator is to distinguish between real and fake samples as much as possible. The objective of the generator is to generate fake samples that the discriminator cannot distinguish. They act gambling with each other.

III. METHOD

A. AVD ARCHITECTURE

This paper proposes a road detection network method based on multi-anti-disturbance, variable-scale spatial context, and soft skip connection. The main body of the structure compares with three parts: detection network, which contains an encoder and a feature decoder, fake feature generator, and a feature discriminator. The whole structure of our network is as shown in Figure 1.

The last layer of activation function is sigmoid. The feature decoder takes non-disturbed features and disturbed features from input images, and the fake feature generator outputs road detection results. A soft skip connection structure is built to bridge the encoder and the decoder for enhancing the fusion of feature information, unlike traditional discriminator, where the fully connected layer is used as the final classifier. The feature discriminator is designed as a fully convolutional network, making inputs of all scales acceptable.

B. OBJECTIVE FUNCTION

The loss function of the detection network (Detector) adopts the Cross-Entropy function and the Smooth L1 loss are shown as follows:

$$L_{CE} = -\frac{1}{n}\sum_i \left[y_i \cdot \log p_i + (1-y_i)\cdot \log(1-p_i)\right] \quad (3)$$

$$L_{1_\text{Smooth}} = \begin{cases} |y_i - p_i| - 0.5 & |y_i - p_i| \leq 1, \\ \dfrac{(y_i - p_i)^2}{2} & |y_i - p_i| > 1. \end{cases} \quad (4)$$

y_i is the label of a sample, whose value is one or zero. p_i is the probability for the prediction without any interferences. From eq. (3)(4), the loss function of the detection network is shown in the following eq. (5), λ_1 is a constant scale factor:

$$L_{\text{Detector}} = L_{CE} + \lambda_1 L_{1_\text{Smooth}} \quad (5)$$

The fake feature generation network is changed to a dual-encoding structure based on the structure of DCGAN. The input of generator G are Gaussian noise z and image x, and multi fake features are G's output; The multi-level fake feature f_G^i are linearly superimposed with the encoder's middle feature layer f_φ^i according to the scale factor α, and the final layer outputs disturbed features. The discriminator

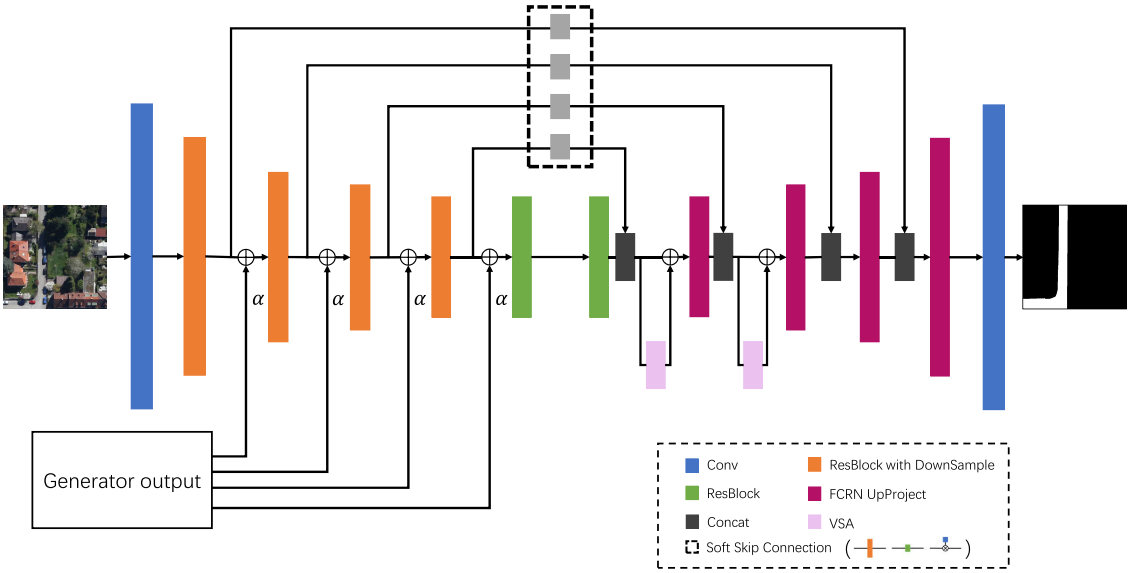

FIGURE 2. The encoder and decoder structure. The proposed architecture is built upon FCRN. The architecture of encoder and decoder: 1. Four soft skip connection links are used between the encoder and the decoder to bridge the low and high layer features, which improve the training efficiency. 2. Fake features from the generator are imposed onto the real features of the encoder. 3. Two variable-scale self-attention (VSA) mechanism modules are introduced to the decoder to extract the spatial context information.

D performs a binary classification task on the image features and disturbed features; the outputs of the real features are one, and the disturbed features are zero so that D can instruct G to generate interferences for the inputs. Therefore, the objective function of the fake feature generation network is shown in the following eq. (8):

$$\varphi\left(x, f_{\varphi}^{i}\right) = f \tag{6}$$

$$G(x, z) = f_{G}^{i} \tag{7}$$

$$L_{GAN}(G, D) = E_{f} \log D(f) + E_{x,z} \log \times \left(1 - D\left(\varphi\left(x, \alpha \cdot G(x, z) + (1-\alpha) f_{\varphi}^{i}\right)\right)\right) \tag{8}$$

To disturb the detection network and form an adversarial relationship, the mean square error interference loss L_D is proposed based on the DCGAN objective function, as shown in the following eq. (10):

$$p_i' = \varphi^{-1}(\alpha \cdot G(x, z) + (1-\alpha)\varphi(x)) \tag{9}$$

$$L_D = -\frac{1}{n} \cdot \sum_{i=1}^{n} \left(y_i - p_i'\right)^2 \tag{10}$$

where p_i' is the probability of correct prediction after adding the interferences, and φ^{-1} stands for the decoder. The objective function of the fake feature generation network, the part of AVD, is shown in the following eq. (11), where λ_2 is a constant.

$$\min_{G} \max_{D} V(G, D) = L_{GAN}(G, D) + \lambda_2 L_D(G) \tag{11}$$

C. DETECTOR ARCHITECTURE IN AVD

The detection network structure of this paper is shown in Figure 2.

The main body of the encoder is composed of five residual blocks, and the decoder combines with five FCRN Fast-Up Projection [5] residual blocks and two variable-scale self-attention (VSA) mechanism modules. The last layer of the decoder uses Sigmoid as the activation function. Adding a VSA mechanism enhances the extraction of spatial context features and improves the detection capability of the detector. In the middle layer of the encoder, we linearly superimpose the output from the fake feature generator according to the scale α. We set the scale factor α to zero during the test. The activation function of the convolutional layer in the residual block adopts LeakyRelu.

D. VARIABLE-SCALE SELF-ATTENTION (VSA)

SAGAN has introduced the self-attention (SA) mechanism to the graphics for the first time and achieves excellent results. However, the original SA mechanism will bring a tremendous amount of calculation when encountering large-size inputs. Therefore, this paper discusses various variable-scale self-attention (VSA) modules; the structure of SA and VSA modules are shown in Figure 3:

The module's input is a feature map, the structures (b) and (d) reduce the dimension and scale of the feature through convolution and output three feature maps. Design (c) only reduces the dimensionality of features through convolution and then performs scale transformation through max-pooling. Then we flatten the last two dimensions of the output feature map. The first two feature maps are multiplied after

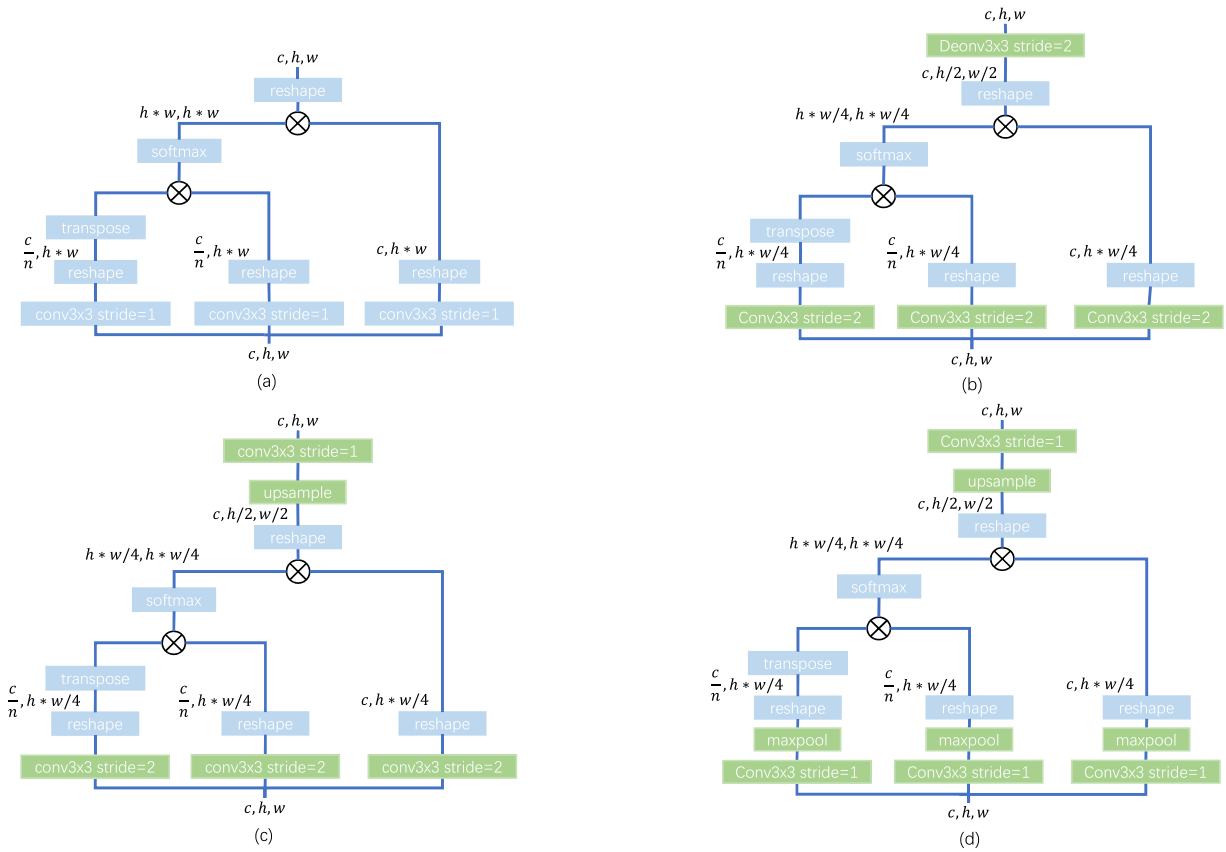

FIGURE 3. Four different structures of the self-attention module: (a) is the original self-attention (SA) module. (b) reduces the self-attention matrix dimension by a larger stride and recovers the scale by a transposed convolution layer. (c) in the bases of (b) uses upsample and convolution instead of transposed convolution. (d) in the bases of (c) uses maxpool and convolution (stride = 1) to replace convolution (stride = 2) for reducing the feature scale.

transpose and passed through the softmax activation function to obtain the attention maps. Multiply the attention map with the last feature map to get self-attention feature maps. (b) and (c) structures restore the original input size by up-sampling and convolution operations. In (d), the layer converts size through deconvolution operation. Through experiments, structure (b) has the best effect, and n is a constant.

E. SOFT SKIP CONNECTION (SSC)

The skip connection increases the shortcut of the output error backpropagation to the underlying features, improving training efficiency. However, it does not consider the incompatibility between cross-layers, which limits the improvement of training accuracy. Therefore, we tried three flexible connection forms, as shown in Figure 4:
where (a) is a 3×3 size convolution kernel; (b) shows a 1×1 size convolution kernel; (c) represents a trainable coefficient gamma. It is found through experiments that the accuracy of structure (b) performs the best.

F. AVD ALGORITHM FLOW

Concatenating image and Gaussian noise together, which fake feature generation network takes as input, and outputs

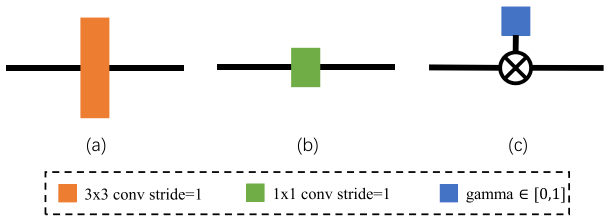

FIGURE 4. Diagram of soft skip connection structure.

fake feature interferences. These disturbances are used as the input of intermediate layers of an encoder, and the discriminator limits the output of the fake feature generator. The fake feature generation network forms an adversarial relationship with the feature discriminator and detection network. The specific algorithm flow is shown in Algorithm 1:

IV. EXPERIMENTS

We implemented our method by PyTorch deep learning framework, the graphics card's model is NVIDIA 2080Ti-11G, and the memory is DDR4-3000-32G. In the following sections, we will introduce our datasets and experimental results.

FIGURE 5. Sample display of Munich remote sensing road and urban road datasets: The first and second rows in the figure are the samples of the Munich remote sensing dataset and the urban road dataset. Then the first and third columns in the figure are the original inputs, and the second and fourth columns are the road labels.

Algorithm 1 Anti-Disturbance Detection Network

Input: image dataset I, target image I_y, and the Gaussian noise z

Output: predict map I_p

1: The sample pictures x_{HR} are randomly selected from the dataset I, and are scaled to obtain the input sample x_{LR}.

2: Take x_{LR} and Gaussian noise z as the input of fake feature generator, and outputs disturbed features of different sizes $f_G^1, f_G^2, f_G^3, f_G^4$, $f_G^i = G(x_{LR}, z)$.

3: The encoder takes input sample x_{LR} and fake features f_G^i as inputs, the encoder's middle layer features and fake features are linearly superimposed according to the proportional coefficient α.

4: The encoder outputs $f_{\varphi mix}$, when $\alpha \neq 0$, $f_{\varphi\,\mathrm{mix}} = \varphi\left(x_{LR}, \alpha f_G^i + (1-\alpha) f_\varphi^i\right)$

5: The encoder outputs f_φ, when $\alpha = 0$, $f_\varphi = \varphi(x_{LR}, f_\varphi^i)$.

6: Feature discriminator needs to distinguish f_φ and $f_{\varphi mix}$, $D(f_\varphi) = 1$, $D(f_{\varphi mix}) = 0$.

7: Encoder φ^{-1} takes disturbed features $f_{\varphi mix}$ as input, and outputs predicted value p, $p = \varphi^{-1}(f_{\varphi mix})$.

8: Update the parameter ω to optimize V.

- *Minimize* $\left\{\nabla \bar{V}(\omega) = \nabla_\omega \left[L_{GAN}\left(\varphi\left(x_{LR}\right), f_G^i, f_\varphi^i\right) + \lambda_1 L_D\left(y, \varphi^{-1}\left(f_{\varphi mix}\right)\right) + \lambda_2 L_{CE}\left(y, \varphi^{-1}\left(f_{\varphi mix}\right)\right) + \lambda_3\left(L_{1_{\mathrm{Smooth}}}\left(y, \varphi^{-1}\left(f_{\mathrm{mix}}\right)\right)\right)\right\}$
- $\omega \leftarrow \omega - \eta \nabla \bar{V}(\omega)$

Repeat steps 1-8 until the model converges or reaches the pre-set number of training rounds, then save the network model.

A. DATASETS

Our experiments use two datasets, the Munich remote sensing road dataset, and the urban dataset. The Munich remote sensing image has a high resolution, but it lacks road calibration. We carry out the calibration work on the road area of the Munich remote sensing image and use it in the experiment of this paper. Currently, the public road dataset lacks night scenes, and night scenes are critical in driving. Many disturbances of lights and shadows in the night road condition bring significant challenges to the detection. So, we collect and calibrate the road pictures in both day and night scenes simultaneously. Some dataset samples are shown in Figure 5. The Munich dataset consists of 20 remote sensing pictures (with a resolution of 5616 × 3744), and we randomly cropped them to obtain 550 samples with a resolution of 512 × 512. The urban road dataset comes from the vehicle's recorder, and we processed the video into pictures and picked up 1100 pictures (with a resolution of 1920 × 1080) containing different road condition information for calibration. The ratio of the number of images between daytime and night is 8:2.

The sample ratio between the train and test is 8:2. We reshaped the image to a resolution of 256 × 256 through the bilinear interpolation. The training adopts Adam optimizer, the encoder and decoder learning rate are set to one ten thousandths. The learning rate of the fake feature generator and feature discriminator is set to five ten thousandths.

B. VISUALIZE ROAD DETECTION RESULTS

We set the superposition coefficient α to 0 during model testing. Using FCN, DeepLab V3, PSPNet, FCRN, and AVD network structures to conduct comparative experiments on the two datasets respectively, the network detection results are shown in Fig. 6 and 7.

From the comparative experiments of Fig.6 and 7, it can be seen that FCN has achieved good results as a pioneer of the full convolution detection network. Compared with FCN, DeepLab has reduced the missed detection part because the introduction of atrous convolution enhances the receptive field. The detection results of PSPNet and DeepLab are relatively close. PSPNet has designed a pyramid pooling structure to complete the fusion of multi-scale features and enhance information extraction but still has false detections when disturbing objects are similar to the road. Compared with the former, FCRN has fewer false detections. This is because the network introduces multiple small-scale convolution kernels

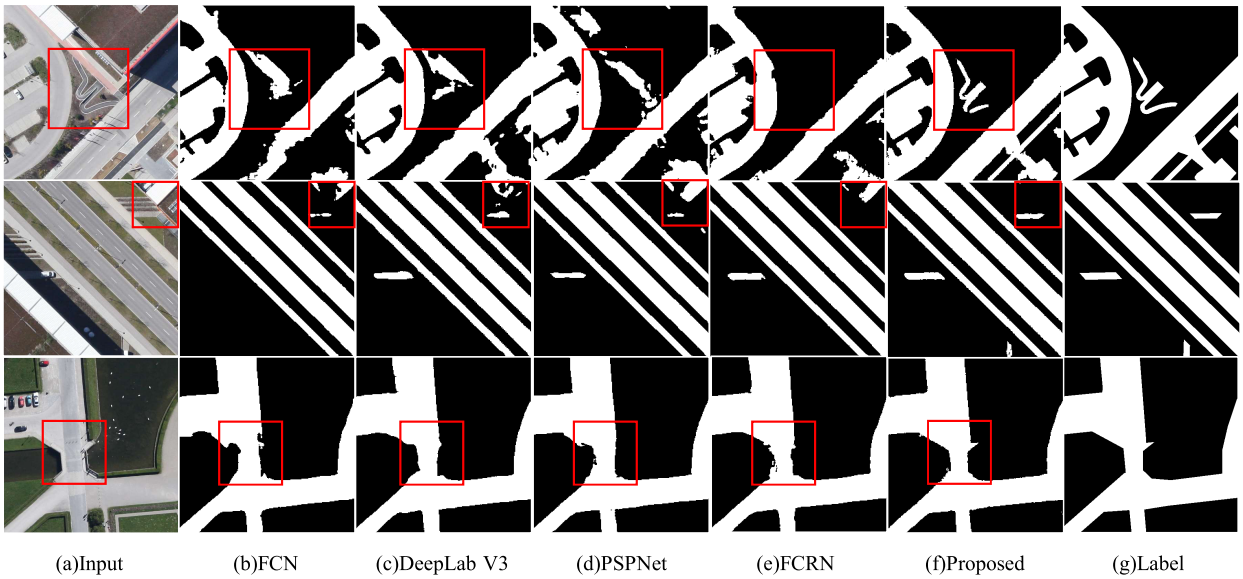

FIGURE 6. Road detection comparison of different methods on Munich dataset. From left to right column: the input sample, the output of FCN, DeepLab V3, and label of the sample. The red frames depict regions of the small pathways, roof, corner. The proposed method outperforms the mainstream methods in these difficult regions.

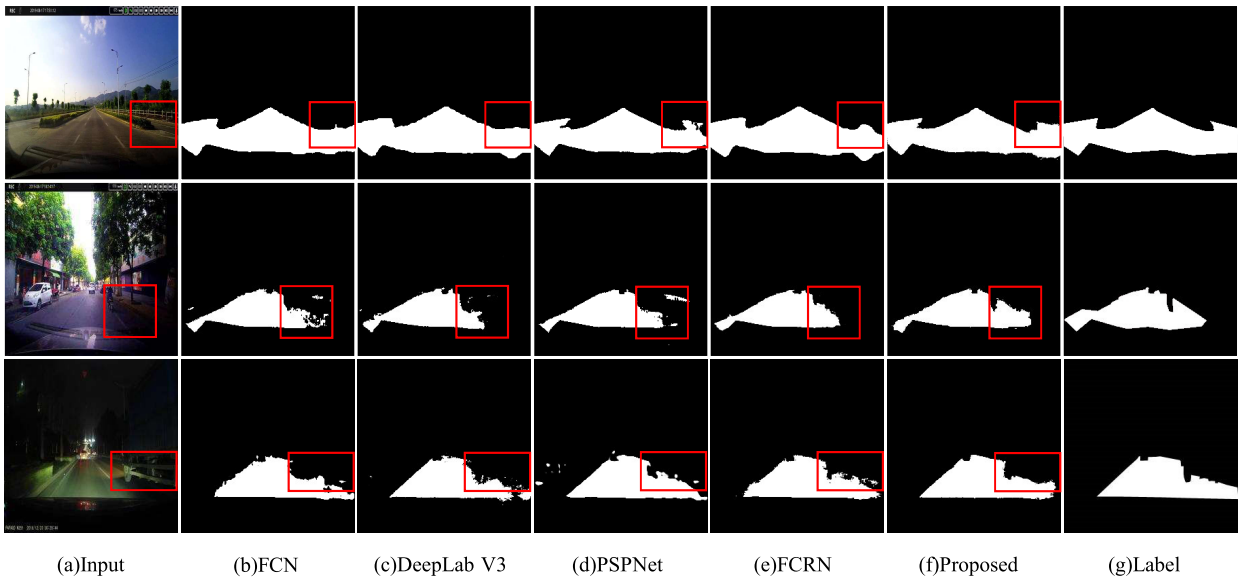

FIGURE 7. Road detection comparison of different methods on urban road dataset. From left to right column: the input sample, the output of FCN, DeepLab V3, and label of the sample. The red frames depict regions of the road corner, cyclist, and contour of the vehicle. The proposed method outperforms the mainstream methods in these marginal regions.

of different sizes to replace large-scale convolutions, reducing the number of parameters and increasing the size of the receptive field. However, FCRN can easily overlook detailed road information. AVD has better anti-disturbance performance than the previous method. The detection result is closest to the real label, and it has smoother edges while suppressing the formation of holes.

C. EVALUATION OF DETECTION RESULTS

In segmentation, Intersection Over Union (IOU) is usually used as a measurement standard for segmentation accuracy, the discrete input points are regarded as positive inspection points in the true value area, and the total length of the true positive inspection points is TP; Discrete points outside the true value area are regarded as false detection points, and the total length of the false positive detection points is FP; the difference between the total sum of all the points marked with true values and TP is the missed value, the missed length FN, the final accuracy IOU equation is as follows eq. (12) shows:

$$\mathrm{IOU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{12}$$

This paper compared the accuracy of AVD with other mainstream models on the road datasets. Table 1 lists the results of the comparison experiments and the computation time of per picture (256 × 256) for each model under the 2080Ti graphic card.

TABLE 1. Comparison of model accuracy on two datasets.

Model	Urban Road IOU/%	Munich IOU/%	Time(ms)
FCN	88.75	69.59	21
FCRN	90.42	75.70	30
DeepLab V3	90.32	73.11	28
PSPNet	90.34	72.52	50
AVD	90.64	77.71	30
AVD (with SSC)	90.72	78.33	32
AVD (with VSA)	90.76	78.65	35
AVD (with SSC and VSA)	**90.80**	**79.06**	38

In the test, we set the scale factor α to 0 and use the detector to overcome the disturbance part alone. It can be seen from the results that our model in this paper has a higher detection accuracy, which is higher than the main-stream segmentation model.

We also conduct a comparative experiment on the influence of multi-level disturbance items on network accuracy. The experimental results are shown in Table 2:

TABLE 2. Comparison of different disturbed features numbers on two datasets.

Disturbance	Urban Road IOU/%	Munich IOU/%
d_3	90.52	77.67
d_2, d_3	**90.64**	**77.71**
d_1, d_2, d_3	88.35	75.86
d_0, d_1, d_2, d_3	81.33	72.42

When the number of disturbance items is 2, the accuracy is the highest. Too few disturbed items will not train the network entirely. However, too many disturbed items will excessively interfere with the network's training, an appropriate number of disturbance items can achieve the optimal result. We conducted comparative experiments on interference features, and tested the influence of the proportion of fake features on the detection accuracy of the detector. The results are shown in Table 3.

TABLE 3. Comparison of different values of α on two datasets.

α	Urban Road IOU/%	Munich IOU/%
0.5	83.23	74.89
0.3	87.15	76.75
0.2	89.40	77.31
0.1	**90.64**	**77.71**
0.05	90.60	77.64
0	90.42	75.70

Table 3 shows that the detection accuracy is the highest when $\alpha = 0.1$, since disturbance is added to high-dimensional features and amplified by the decoder, it will produce a better intervention effect. Therefore, adding disturbance appropriately from the generator is beneficial to improve the network's ability to extract features. We compare the impact of SA and our proposed VSA structure on the detection accuracy of the detector. The results are shown in Table 4.

TABLE 4. Comparison of the different structure of self-attention on two datasets.

Name	Urban Road IOU/%	Munich IOU/%
SA(a)	90.70	78.09
VSA(b)	90.71	78.22
VSA(c)	**90.76**	**78.65**
VSA(d)	90.73	78.34

Compared with SA, VSA introduces more and larger convolution kernels. Due to the increase of trainable parameters, the association of each pixel information on the feature map is enhanced. Finally, we conducted comparative experiments on the improved three types of soft skip connection (SSC). The experimental results are shown in Table 5:

TABLE 5. Comparison of the different structure of skip connection on two datasets.

Name	Urban Road IOU/%	Munich IOU/%
SC(a)	90.78	78.68
SSC(b)	90.79	79.01
SSC(c)	**90.80**	**79.06**
SSC(d)	90.77	78.66

The structure of SSC(b) is a 3 × 3 convolution kernel. The design of SSC(c) is a 1 × 1 convolution kernel; the structure of SSC(d) is a trainable scale factor. Experiments have proved that using a 1 × 1 size convolution kernel for skip connection can achieve higher detection accuracy. For a 1 × 1 size, the convolution kernel can not only better enrich the feature information but also get the best error distribution, and propagation relative to the high-level error of the skip connection must be returned to the bottom-level way.

V. CONCLUSION

In this paper, we present a novel detection network structure based on multi-anti-disturbance, variable-scale spatial context and soft skip connection. By overlapping generated fake feature onto the middle layer of the encoder and introducing the variable-scale self-attention module and soft skip connection into the decoder, this method improves the detection network's anti-disturbance ability. Comparing with several mainstream networks (such as FCN, FCRN, PSPNet, DeepLab V3, etc.), on the datasets of Munich remote sensing and urban road, the proposed method improves the road detection accuracy by 3%, 0.4%, obviously show the effect and efficiency of our method. In future work, we will proceed more researches on the disturbance patterns, ranges, generation methods and etc., to mimic the real disturbances well.

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